Machine Learning-Based Drug Repurposing Discovery for Disordered (Non-Globular) MYC and CTNNB1 Oncoproteins

Computer *Engineering Senior Design Project*

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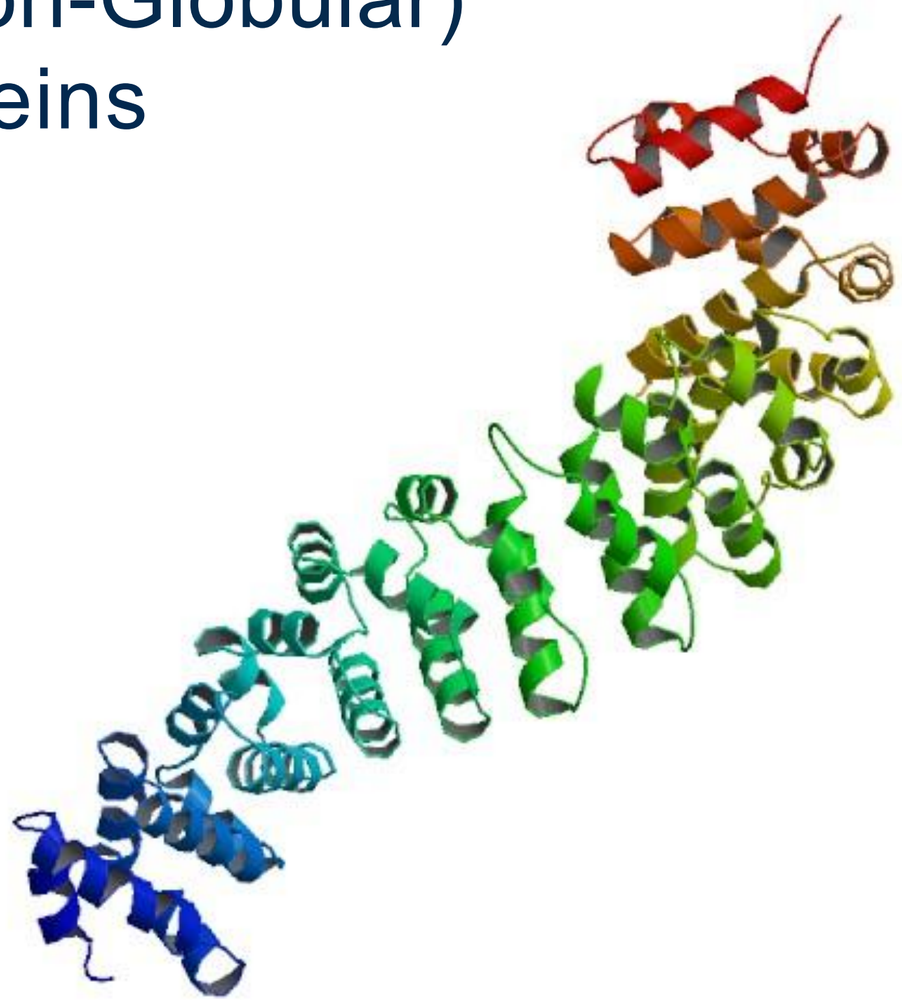
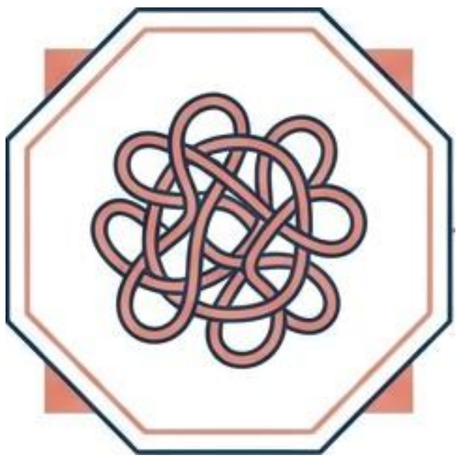


Summary

An innovative computational approach to bridge the gap for 'undruggable' cancer targets.

The Problem



The Solution

The Process

The Impact

The Problem

MYC & CTNNB1 are 'undruggable' IDPs lacking stable pockets

The Solution

ML strategy using ChEMBL bioactivity data (RF & GNN models)

The Process

Screening PubChem library & validating via 100ns MD Simulations

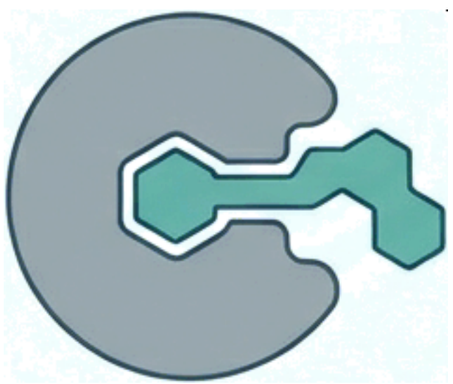
The Impact

Innovative computational method & list of validated candidate molecules

Introduction: The ‘Undruggable’ Challenge

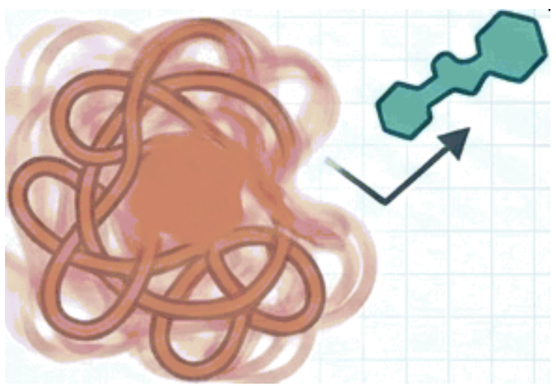
Structural Failure Modes

Classical “Lock and Key”



Stable binding pockets allow traditional drugs to lock in.

The IDP Reality (MYC & ß-catenin)



Conformational heterogeneity prevents stable binding.

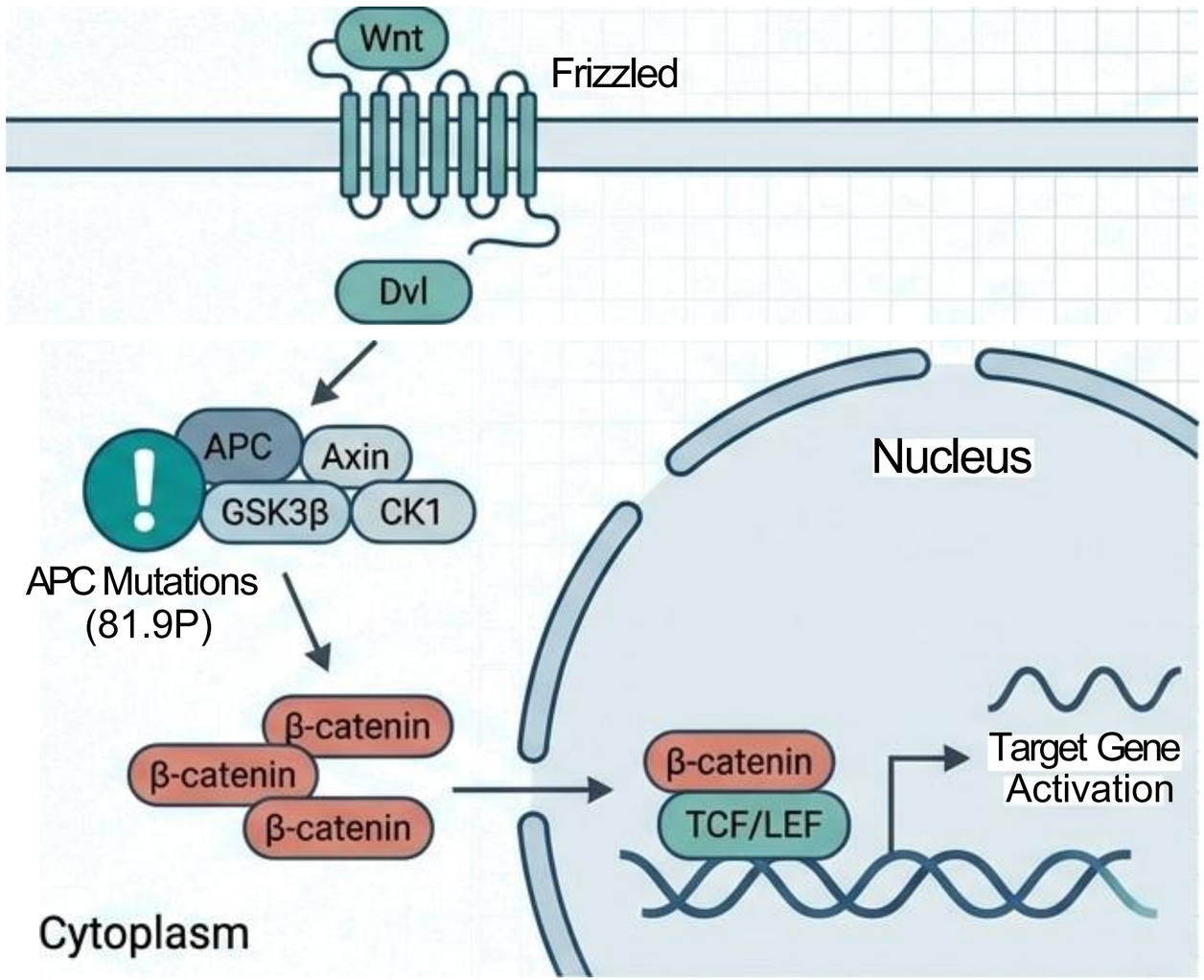
Why Methods Fail: Traditional structure-based drug design relies on static shapes. IDPs are dynamic, lacking the deep pockets required for this approach. Current research focuses on characterization, leaving a gap in ligand binding prediction.

Introduction: The Cancer Targets

The Stakes: Cancer is a leading global health problem. Identifying molecular treatment targets is critical.

Target 1: MYC
- A transcription factor central to cell proliferation and metabolism.
- Overactivation is a hallmark of many cancers.

Target 2: CTNNB1 (ß-catenin)
- Key effector of the Wnt/ß-catenin signaling pathway.
- Key Statistic: APC mutations found in
- 81.9% of colorectal cancers; direct
- CTNNB1 mutations in 48%.



Research Question

Can potential inhibitors targeting disordered MYC and CTNNB1 oncoproteins be predicted using machine learning models based on bioactivity data?

Shifting the paradigm from structure-based search

to data-driven bioactivity prediction.

- This research was developed within the scope of the TÜBİTAK 2209-A Undergraduate Students Research Projects Support Program for the 2025 first-period application.

Hypothesis

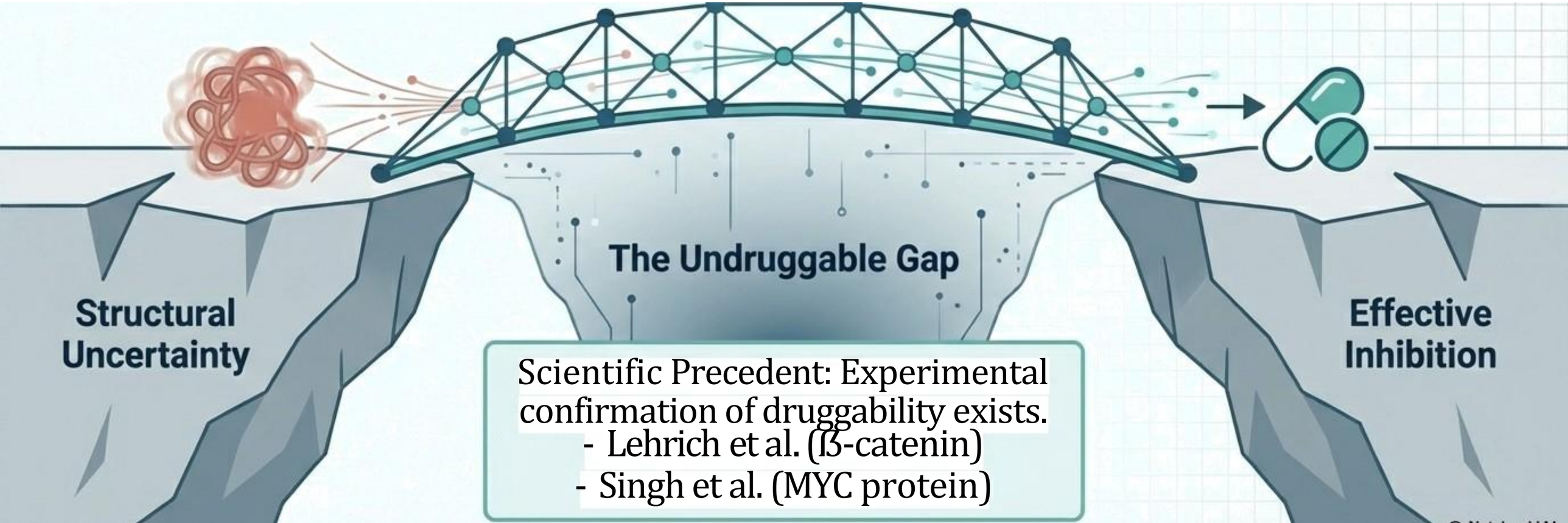
The Proposition

ML models trained on large-scale bioactivity data can identify binders independent of protein shape.

Bioactivity-Based Machine Learning

The Goal

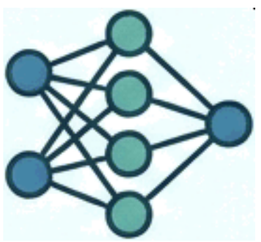
Demonstrate applicability of ML strategies for MYC and ß-catenin.



Scientific Precedent: Experimental confirmation of druggability exists.
- Lehrich et al. (ß-catenin)
- Singh et al. (MYC protein)

Concrete Objectives

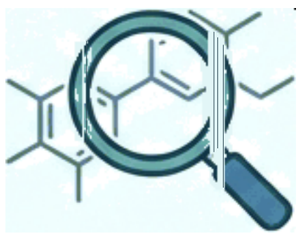
Develop



Develop high—accuracy classification models (Random Forest, Gradlent Boosting, GNN) using data from ChEMBL, PubChem, and BindingDB (>1o,ooo,ooo molecules).

Target Performance: AUC ≥ 0.95

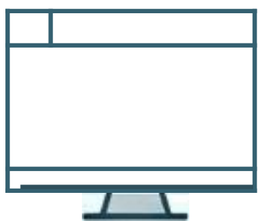
Identify



Screen the PubChem BioAssay database to find hit molecules with low toxicity that can be repurposed for MYC-MAX and §-catenin-TCFJ interactions.

Target: Hit Molecules with Low Toxicity

Validate

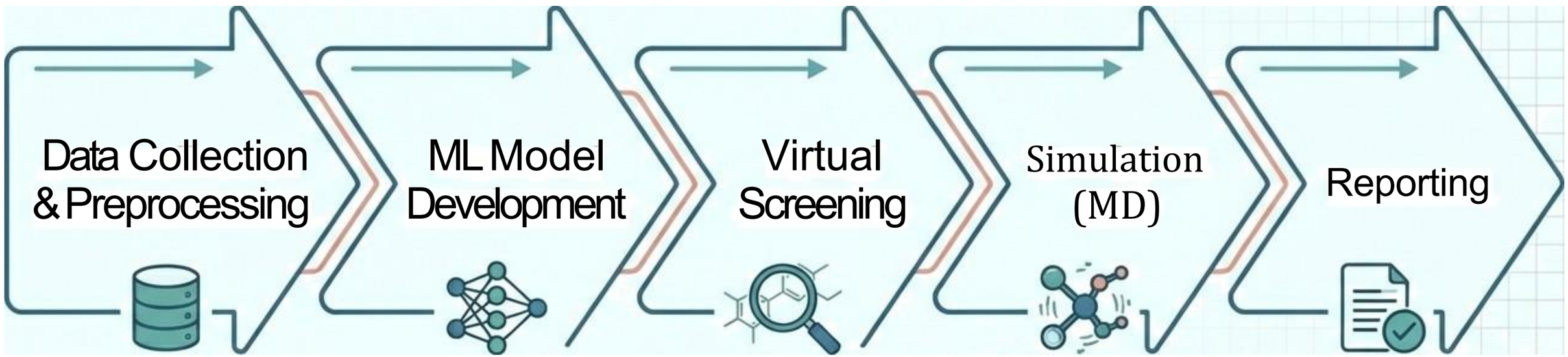


Computationally validate binding affinities of the top hits (nssJ via Molecular docking (AutoDockVina), MD simulations (loo ns), and MM-GBSA calculations.

Target: Validated Stability

Methods: Workflow Overview

Based on QSAR-ML workflows (Sharma & Arumugam) and Bayesian ML (Vignaux et a1.)



ChEMBL, PubChem, BindingDB

RF, GBM, GNN Training

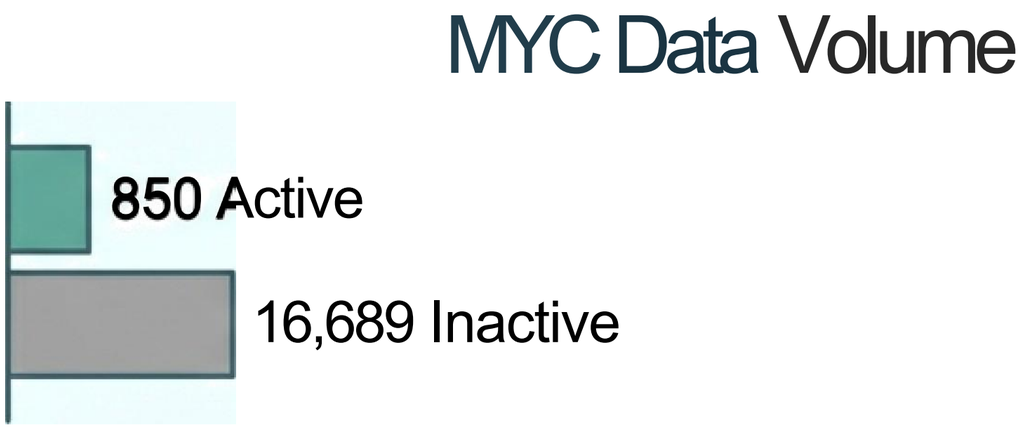
PubChem BioAssay Filtering

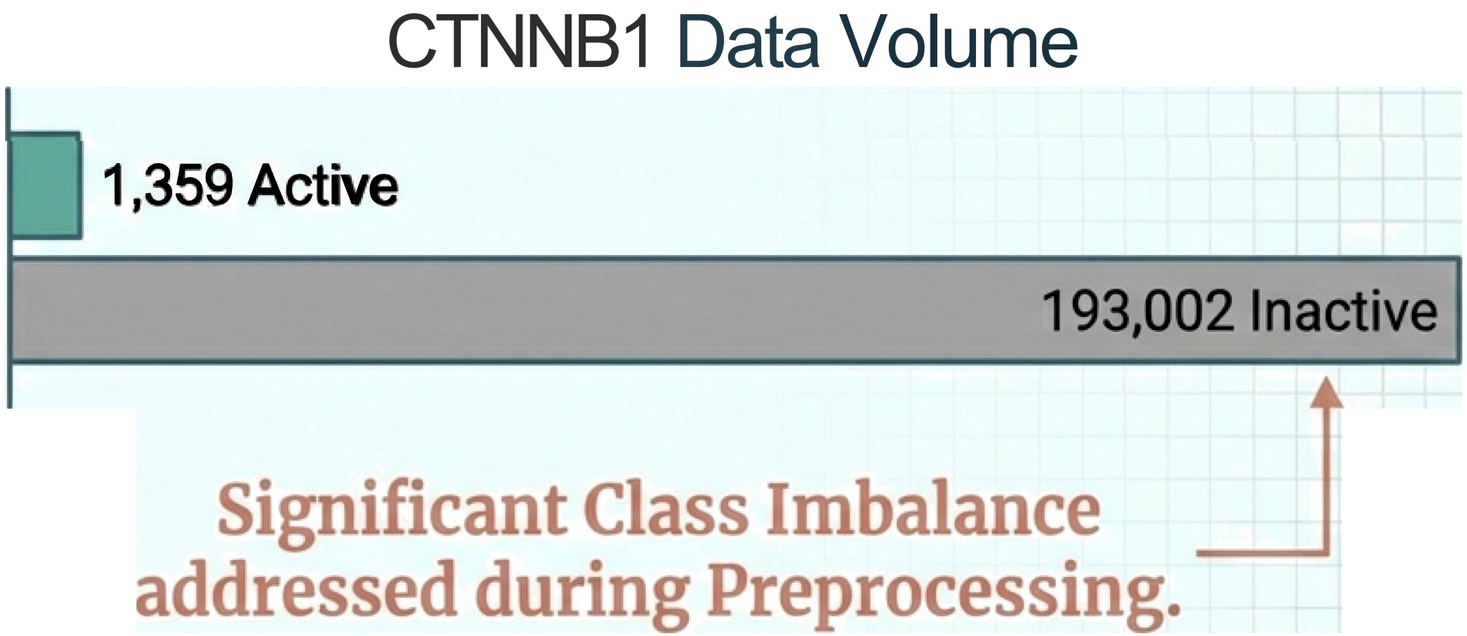
100ns Validation

Candidate List & Paper

2.1 Data Collection and Preprocessing

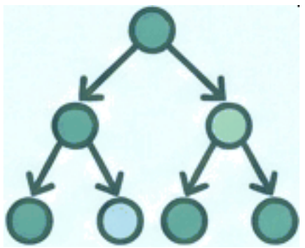
- Sources: ChEMBL, PubChem, and BindingDB.
- Target Interactions: MYC-M4X and 9 catenin-TCF4



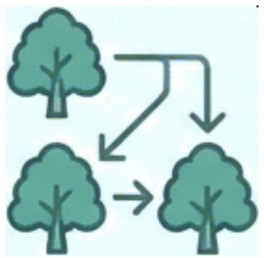


2.2 Machine Learning Model Development and Optimization

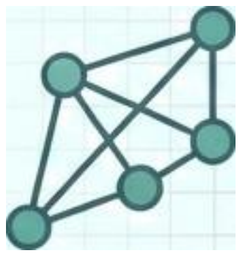
Algorithms



Random Forest (RF)



Gradient Boosting Machines (GBM)



Graph Neural Networks (GNN)

Optimization & Metrics

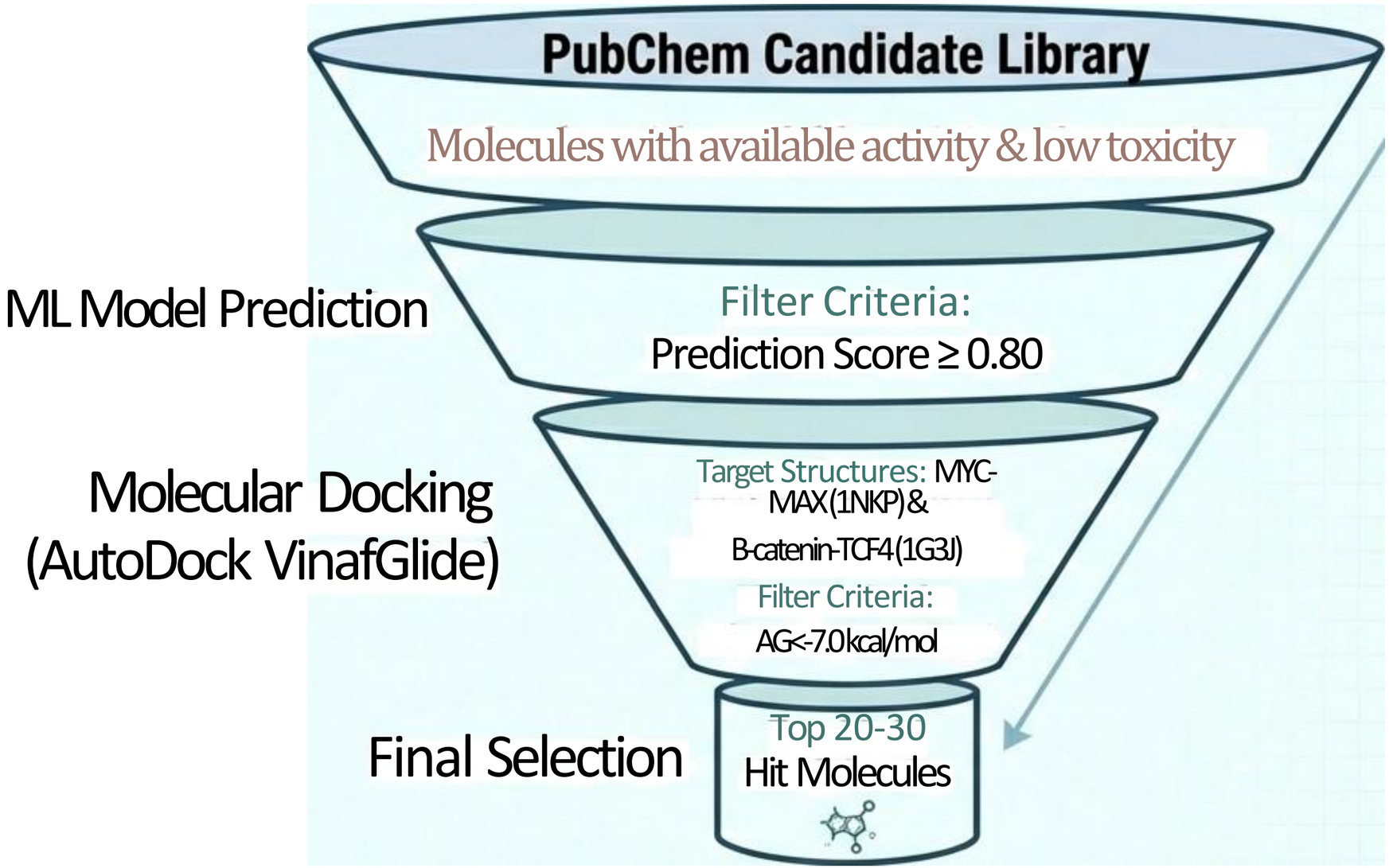
Optimization Strategy:
Hyperparameter optimization via Grid Search and Randomized Search.

Validation Metrics:
10-fold cross-validation evaluating AUC, precision, recall, and F1-score.

Benchmarks

Literature Benchmarks:
- Random Forest: 98% AUC (Sharma & Arumugam)
- Graph Neural Networks: 99% Accuracy vs 82% traditional (Noor et al.)

2.3 Virtual Screening and Hit Molecule Identification



2.4 Molecular Dynamics Simulations

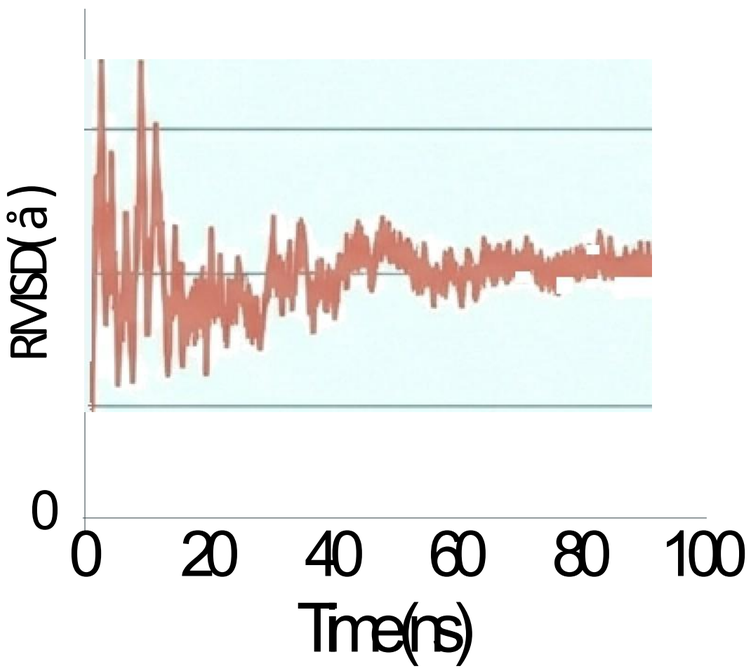
Validating physical stability in a 100ns simulation environment.

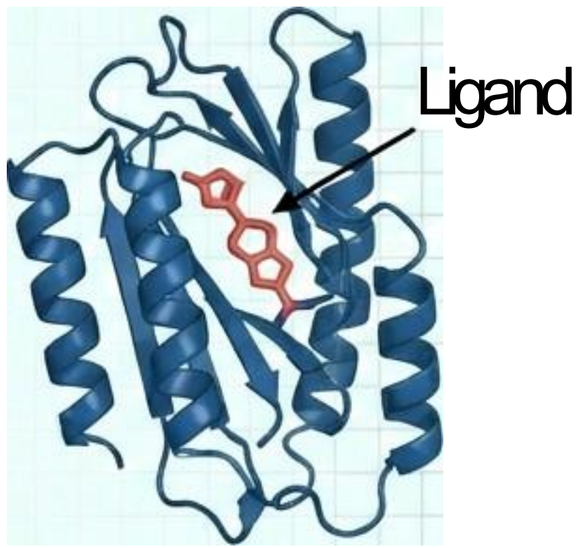
The Setup

- Soflware: GROMACS or AMBER
- Duration: 1o0 ns
- Metrics: RMSD (Root Mean Square Deviation), RMSF, Hydrogen Bond Formation.

The Visualization

RMSD Analysis





Protein-Ligand Complex Stability.

Results& Deliverables

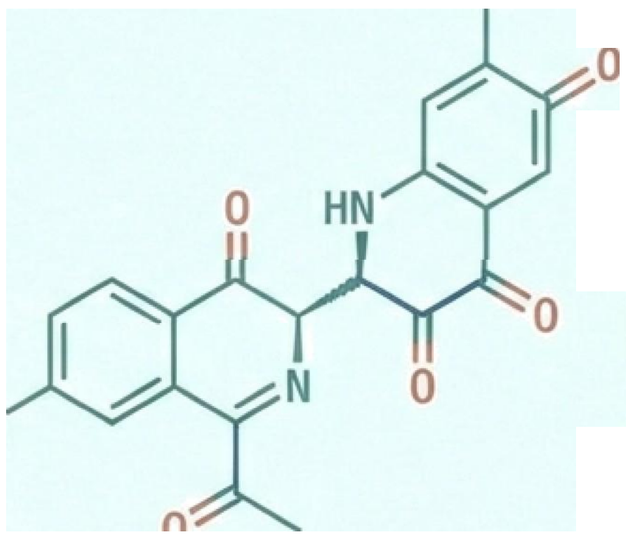
Primary Deliverable:

A comprehensive project report containing a prioritized list of candidate molecules.

Dissemination:

Report conversion into a conference paper article.

Drug Candidate Data Card



ML Prediction Score: 0.96

Docking Score: -8.2 kcal/mol

ADMET Profile: Low Toxicity

MD Stability: Stable at 100ns

Future Plans

- **Conference Submission & International Visibility:** One of our immediate targets is to prepare an abstract based on our results for the ML4NGP (Machine Learning for Non-Globular Proteins) COST Action conference. Since my advisor is an active working group member of this network, this will be a strategic opportunity to present our findings as a poster or oral presentation to an international audience

- **Web Visualization & Data Sharing:** To increase the accessibility of our research, we plan to develop a web-based visualization platform. This tool will allow other researchers to interact with our 100 ns molecular dynamics results and explore the chemical properties of our identified hit molecules

Key References

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- Cui et al. (2018). Is ß-Catenin a Druggable Target for Cancer Therapy? Trends in Biochemical Sciences.
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- Vignaux et al. (2020). Machine Learning for Discovery of GSK3ß Inhibitors. ACS Omega.

Thanks..